TableTask ECG-to-HR Workflow Guide

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Workflow through Step 3: metadata/QC, ECG processing, pretrial QC, and trial segmentation

Prepared for research assistants joining the TableTask Movesense analysis workflow.

Scope: this guide documents the pipeline through Step 3 only. Step 4 and Step 5 are planned downstream analysis steps.

Key rule: every derived result must remain traceable to the script, input files, and QC decision that produced it.

1. Current folder structure and scripts

The current project folder is organized as follows. The names below match the working folder shown in Finder.

```
TableTask/
  processed_ecg_hr/
    trial_segments_4hz/
    pretrial_2min_qc_v2/
    plots/
      _organized_by_dyad_preview/
      _organized_by_sensor/
      hr_before_after_correction/
      hr_time_series/
      hrv_psd/
      qc_summary/
      raw_vs_clean/
      rpeaks/
      dyad_session_preview_not_for_final_analysis/
      poincare/
    hr_interpolated_4hz/
    hr_interpolated_4hz_original/
    hrv_metrics/
    qc/
    rpeak_rr_corrected/
    rpeak_rr_original/
  qc_outputs/
  T24Z8B~S/
  Table MoveSense Sensor/
  aggregated_data.csv
  table_timestamps.csv
  Table Task 2026.xlsx
  01_script_prepare_tabletask_metadata_qc_EG_final.py
  02_script_process_ecg_to_hr_EG_final.py
  02a_script_organize_step2_plots_by_sensor_EG.py
  02b_script_extract_pretrial_2min_qc_EG_v2_final.py
  02c_script_organize_step2b_pretrial_qc_plots_by_sensor_EG_final.py
  03_script_segment_hr_by_trial_EG.py
```

File/folder descriptions:

Item	Purpose
Table MoveSense Sensor/	Raw participant-worn Movesense session folders. Contains ECG, accelerometer, Euler angles, and annotations.
T24Z8B~S/	Raw table accelerometer folder. Step 1 does not use this directly once table_timestamps.csv exists.
table_timestamps.csv	Table-motion timing windows derived from the table accelerometer. Used to define practice/task windows.
Table Task 2026.xlsx	Metadata: participant IDs, A/B roles, sensor suffixes, trial order, pilot status, and comments.
aggregated_data.csv	Trial-level behavioural and subjective-rating summaries.
qc_outputs/	Step 1 outputs. Most important file: veronica_analysis_units.csv.
processed_ecg_hr/	Step 2, Step 2b, and Step 3 outputs.

Important processed_ecg_hr/ subfolders: hr_interpolated_4hz/ contains the corrected 4 Hz HR files used downstream; pretrial_2min_qc_v2/ contains Step 2b QC; trial_segments_4hz/ contains Step 3 paired trial/window HR files; plots/ contains Step 2 visual QC plots, including _organized_by_sensor and _organized_by_dyad_preview.

2. Overall workflow

The analysis through Step 3 is shown below. Step 3 is not a synchrony analysis step. It only creates paired trial-level HR files and carries QC labels forward.

Complete Updated Pipeline Workflow:

[insert flowchart]

Figure 1. Overall TableTask ECG-to-HR workflow through Step 3. Step 3 receives three inputs: Step 1 timing/metadata, Step 2 corrected HR files, and Step 2b pretrial QC flags.

Step	Purpose	Script(s)	Main inputs	Main outputs	Lena pipeline relationship
Step 1	Build the master metadata/QC analysis-unit table.	01_script_prepare_tabletask_metadata_qc_EG_final.py	Table MoveSense Sensor/; Table Task 2026.xlsx; table_timestamps.csv; aggregated_data.csv	qc_outputs/TableTask_Master_QC.xlsx; qc_outputs/veronica_analysis_units.csv/xlsx; metadata_qc_summary.txt	New TableTask-specific step.
Step 2	Process raw ECG to cleaned ECG, corrected R-peaks/RR, corrected 4 Hz HR, and QC plots.	02_script_process_ecg_to_hr_EG_final.py; 02a_script_organize_step2_plots_by_sensor_EG.py	Raw Movesense ECG files; qc_outputs/veronica_analysis_units.csv	processed_ecg_hr/hr_interpolated_4hz/; rpeak_rr_corrected/; hrv_metrics/; plots/; qc/	Adapted from Lena's ECG-processing workflow. The file organization and TableTask metadata integration are project-specific.
Step 2b	Extract first 2 minutes of each recording to assess pretask sensor quality.	02b_script_extract_pretrial_2min_qc_EG_v2_final.py; 02c_script_organize_step2b_pretrial_qc_plots_by_sensor_EG_final.py	processed_ecg_hr/hr_interpolated_4hz/; rpeak_rr_corrected/; Step 1 metadata	processed_ecg_hr/pretrial_2min_qc_v2/qc/ and plots/	Veronica-added QC extension, not from Lena's original pipeline.
Step 3	Cut corrected full-session HR into table-task/practice windows and align A/B HR on a common 4 Hz grid.	03_script_segment_hr_by_trial_EG.py	qc_outputs/veronica_analysis_units.csv; processed_ecg_hr/hr_interpolated_4hz/; pretrial_2min_qc_v2 QC files	processed_ecg_hr/trial_segments_4hz/paired_trial_hr/ and qc/	Project-specific segmentation step. It uses Lena-style HR outputs but TableTask-specific timing and conditions.

3. Step 1 - metadata, timing, and master QC file

Field	Content
Purpose	Create the master dyad-window table that links each recording folder, sensors, participants, condition, practice/trial window, timing window, behavioural ratings, and exclusion flags.
Script used	01_script_prepare_tabletask_metadata_qc_EG_final.py
Main inputs	Table MoveSense Sensor/, Table Task 2026.xlsx, table_timestamps.csv, aggregated_data.csv
Main outputs	qc_outputs/TableTask_Master_QC.xlsx, qc_outputs/veronica_analysis_units.csv, qc_outputs/veronica_analysis_units.xlsx, qc_outputs/metadata_qc_summary.txt
Use of master QC file	Defines analysis units, condition labels, candidate windows, trial numbers, accel_start_unix, accel_end_unix, usable_for_dyadic_ecg, exclude_from_main_analysis, and behavioural/rating variables. Step 3 uses this file as the source of trial timing.

Step 1 verified counts

- 162 analysis-unit rows: 18 dyad sessions x 9 candidate windows.
- 156 rows usable for dyadic ECG segmentation.
- 6 rows unusable for dyadic ECG because both sensors did not fully cover the table-motion window.
- 115 rows are main-analysis eligible real task trials after Step 1 exclusions.
- Candidate window 1 is practice. Candidate windows 2-9 are real trials 1-8.

Why the master QC file matters

- It prevents manual matching errors between folder names, sensor IDs, dyad IDs, trial labels, and condition order.
- It preserves pilot, practice, missing-rating, and known-session comments instead of dropping them silently.
- It allows later scripts to use explicit inclusion/exclusion fields rather than hard-coded assumptions.

4. Step 2 - raw ECG to corrected 4 Hz HR

Field	Content
Purpose	Convert each participant sensor recording from raw ECG to corrected R-peaks, RR intervals, corrected 4 Hz HR time series, HRV outputs, and visual QC plots.
Script used	02_script_process_ecg_to_hr_EG_final.py
Plot organizer	02a_script_organize_step2_plots_by_sensor_EG.py
Pipeline origin	This step is adapted from Lena's ECG-processing workflow: ECG cleaning, R-peak detection, R-peak correction, RR intervals, interpolated HR, HRV, and QC plots. The TableTask version adapts file discovery, metadata joins, and outputs to this dataset.
Main inputs	Table MoveSense Sensor/, qc_outputs/veronica_analysis_units.csv
Main outputs	processed_ecg_hr/hr_interpolated_4hz/, rpeak_rr_corrected/, hrv_metrics/, plots/, qc/
Final run status	36/36 sensor recordings processed successfully; 252 Step 2 plot files generated. Full cleaned ECG CSV files were not retained because downstream steps use corrected 4 Hz HR and RR/R-peak outputs.

Step 2 QC criteria and why they were selected

Criterion	Why it was used
ECG pattern/shape	Good signal shows repeated recognizable QRS/R-peak structure with a reasonably stable baseline. This is checked because HR accuracy depends on detecting real heartbeats, not movement spikes.
R-peak placement	Corrected R-peak markers should sit on the main cardiac peaks. If R-peaks are placed on noise, all later RR and HR values become unreliable.
HR after correction	Corrected HR should be mostly physiologically plausible and should not remain dominated by dense spikes or impossible drops. Correction can reduce outliers, but it cannot rescue fundamentally poor ECG.
Poincare shape	A usable RR Poincare plot should show an organized diagonal, cigar-shaped, or comet-shaped pattern with limited extreme outliers. Random scatter or points spread across thousands of milliseconds indicate artifact-driven RR intervals.
Dyad preview	Whole-session dyad previews are diagnostic only. They show whether both participants have plausible HR traces but are not final FF/FE synchrony results.

Step 2 visual QC lists

These are visual Step 2 QC categories from the full-session plots. They are used for review not as automatic final exclusions. Pilot sessions are marked because they remain excluded from primary analysis even if visually acceptable.

Clean/acceptable sensors (14)	Caution sensors (8)	Poor/problematic sensors (14)
2026_04_07-23_37_18 / 242930002208 / role B 2026_04_07-23_37_18 / 242930002204 / role A 2026_03_31-22_33_41 / 242930002208 / role A 2026_03_31-22_33_41 / 242930002204 / role B 2026_03_27-17_43_17 / 242930002208 / role B 2026_03_27-17_43_17 / 242930002204 / role A 2026_03_27-18_34_14 / 242930002208 / role A 2026_03_09-18_34_03 / 242930002204 / role B 2026_03_09-18_34_03 / 242930002208 / role A 2026_03_10-18_36_10 / 242930002204 / role A (pilot) 2026_03_10-18_36_10 / 242930002208 / role B (pilot) 2026_02_26-19_47_35 / 242930002204 / role A (pilot) 2026_02_26-19_47_35 / 242930002208 / role B (pilot) 2026_03_03-19_31_30 / 251430002238 / role A (pilot)	2026_03_31-23_35_29 / 242930002208 / role B 2026_03_31-23_35_29 / 242930002204 / role A 2026_03_31-21_32_11 / 242930002204 / role B 2026_03_17-22_42_37 / 242930002204 / role B 2026_03_17-22_42_37 / 242930002208 / role A 2026_03_03-22_32_43 / 242930002204 / role A 2026_03_26-19_41_35 / 251430002277 / role A 2026_03_26-19_41_35 / 242930002205 / role B	2026_04_07-17_29_26 / 242930002204 / role A 2026_04_07-17_29_26 / 242930002208 / role B 2026_04_07-22_33_07 / 242930002208 / role A 2026_04_07-22_33_07 / 242930002204 / role B 2026_03_31-18_39_54 / 251430002239 / role B 2026_03_31-18_39_54 / 242930002205 / role A 2026_03_31-21_32_11 / 242930002208 / role A 2026_03_27-18_34_14 / 242930002204 / role B 2026_03_26-18_28_36 / 251430002277 / role A 2026_03_26-18_28_36 / 251430002239 / role B 2026_03_03-19_31_30 / 251430002247 / role B (pilot) 2026_03_03-22_32_43 / 242930002208 / role B 2026_03_03-23_45_18 / 242930002208 / role A 2026_03_03-23_45_18 / 242930002204 / role B

Clean dyads (6)	Caution dyads (3)	Poor/problematic dyads (9)
2026_04_07-23_37_18 / P41_P42 2026_03_31-22_33_41 / P29_P30 2026_03_27-17_43_17 / P21_P22 2026_03_09-18_34_03 / P09_P10 2026_03_10-18_36_10 / P11_P12 (pilot) 2026_02_26-19_47_35 / P01_P02 (pilot)	2026_03_31-23_35_29 / P31_P32 2026_03_17-22_42_37 / P15_P16 2026_03_26-19_41_35 / P19_P20	2026_04_07-22_33_07 / P39_P40 2026_04_07-17_29_26 / P33_P34 2026_03_31-21_32_11 / P27_P28 2026_03_31-18_39_54 / P25_P26 2026_03_27-18_34_14 / P23_P24 2026_03_26-18_28_36 / P17_P18 2026_03_03-23_45_18 / P07_P08 2026_03_03-22_32_43 / P05_P06 2026_03_03-19_31_30 /

		P03_P04 (pilot)

5. Step 2 example plots

The same clean session/sensor is used across Step 2 and Step 2b where applicable: 2026_04_07-23_37_18, sensor 242930002204, role A. This makes the interpretation easier for new research assistants.

Step 2 plot examples: clean versus poor

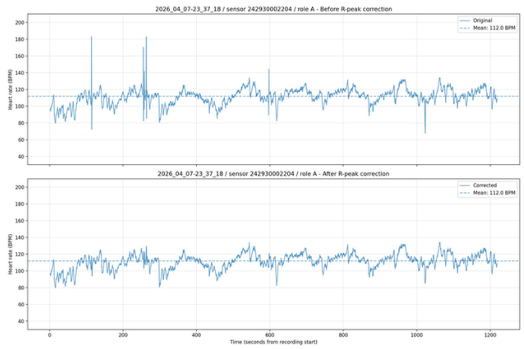
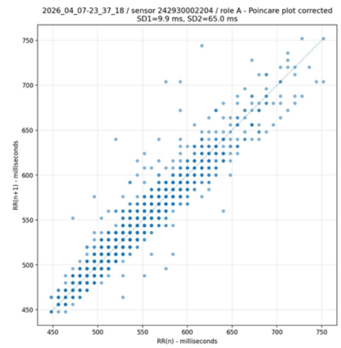
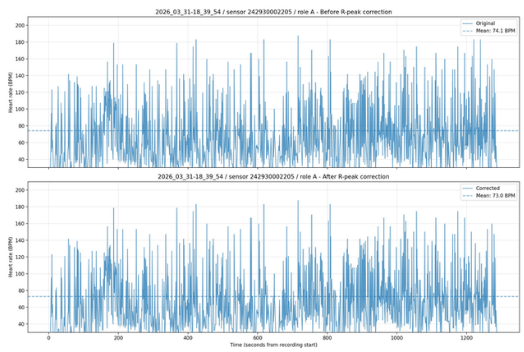
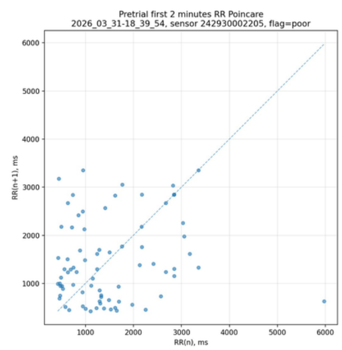
Clean Step 2 HR: 2026_04_07-23_37_18 / 242930002204	Clean Step 2 Poincare: diagonal/comet-shaped RR structure
	
Poor Step 2 HR: 2026_03_31-18_39_54 / 242930002205	Poor RR structure shown for contrast: dispersed intervals
	

Figure 2. Step 2 examples comparing clean and poor sensor recordings. Clean example uses 2026_04_07-23_37_18 / sensor 242930002204 / role A. Poor example uses 2026_03_31-18_39_54 / sensor 242930002205 / role A. The clean Poincare pattern is organized and diagonal/comet-shaped; the poor pattern is scattered and artifact-risky.

6. Step 2b and 2c - first 2-minute pretrial QC

Field	Content
Purpose	Assess signal quality during the first two minutes of each recording, before the table-carrying task. This was added after Veronica's feedback to distinguish baseline sensor-quality problems from task-movement artifact.
Step 2b script	02b_script_extract_pretrial_2min_qc_EG_v2_final.py
Step 2c script	02c_script_organize_step2b_pretrial_qc_plots_by_sensor_EG_final.py
Pipeline origin	This is a new Veronica/TableTask QC step. It is not a direct Lena pipeline step.
Main inputs	processed_ecg_hr/hr_interpolated_4hz/, rpeak_rr_corrected/, qc/02_ecg_to_hr_processing_summary.csv, raw ECG when ECG zoom plots are generated
Main outputs	processed_ecg_hr/pretrial_2min_qc_v2/qc/ and processed_ecg_hr/pretrial_2min_qc_v2/plots/_organized_by_sensor/

Step 2b objective QC thresholds and why they were selected

Flag	Decision rule	Why this matters
Clean	Assigned only if no caution or poor criterion is triggered.	Indicates the first 2 min has sufficient coverage, plausible HR/RR, and stable enough HR to use as a clean pretrial signal-quality reference.
Caution	Triggered by intermediate issues: 60-100 s HR duration, coverage below 80%, RR plausible 90-95%, extreme HR >0.5% and <=5%, min HR 30-40 BPM, max HR 200-220 BPM, HR SD 20-30 BPM, or HR range 100-150 BPM.	Marks recordings that are not clearly failed but need visual review. These can sometimes be usable in later trial windows.
Poor	Triggered by any severe issue: HR duration <60 s, fewer than 30 RR intervals, RR plausible <90%, extreme HR >5%, min HR <30 BPM, max HR >220 BPM, HR SD >30 BPM, or HR range >150 BPM.	Marks high-risk recordings where the signal is likely unreliable before task movement. Poor overrides caution.

Step 2b QC lists

Clean sensors (21)	Caution sensors (3)	Poor sensors (12)
2026_04_07-23_37_18 / 242930002208 / role B 2026_04_07-23_37_18 / 242930002204 / role A 2026_03_31-23_35_29 / 242930002208 / role B 2026_03_31-23_35_29 / 242930002204 / role A 2026_03_31-22_33_41 / 242930002208 / role A 2026_03_31-22_33_41 / 242930002204 / role B 2026_03_31-21_32_11 / 242930002204 / role B 2026_03_27-18_34_14 / 242930002208 / role A 2026_03_27-17_43_17 /	2026_04_07-22_33_07 / 242930002204 / role B 2026_04_07-17_29_26 / 242930002204 / role A 2026_03_17-22_42_37 / 242930002208 / role A	2026_04_07-22_33_07 / 242930002208 / role A 2026_04_07-17_29_26 / 242930002208 / role B 2026_03_31-21_32_11 / 242930002208 / role A 2026_03_31-18_39_54 / 251430002239 / role B 2026_03_31-18_39_54 / 242930002205 / role A 2026_03_27-18_34_14 / 242930002204 / role B 2026_03_26-18_28_36 / 251430002277 / role A 2026_03_26-18_28_36 / 251430002239 / role B 2026_03_03-23_45_18 /

242930002208 / role B 2026_03_27-17_43_17 / 242930002204 / role A 2026_03_26-19_41_35 / 251430002277 / role A 2026_03_26-19_41_35 / 242930002205 / role B 2026_03_17-22_42_37 / 242930002204 / role B 2026_03_10-18_36_10 / 242930002208 / role B (pilot) 2026_03_10-18_36_10 / 242930002204 / role A (pilot) 2026_03_09-18_34_03 / 242930002208 / role A 2026_03_09-18_34_03 / 242930002204 / role B 2026_03_03-22_32_43 / 242930002204 / role A 2026_03_03-19_31_30 / 251430002238 / role A (pilot) 2026_02_26-19_47_35 / 242930002208 / role B (pilot) 2026_02_26-19_47_35 / 242930002204 / role A (pilot)		242930002208 / role A 2026_03_03-23_45_18 / 242930002204 / role B 2026_03_03-22_32_43 / 242930002208 / role B 2026_03_03-19_31_30 / 251430002247 / role B (pilot)

Both-clean dyads (8)	**Caution dyads (1)**	**Poor dyads (9)**
2026_04_07-23_37_18 / P41_P42 2026_03_31-23_35_29 / P31_P32 2026_03_31-22_33_41 / P29_P30 2026_03_27-17_43_17 / P21_P22 2026_03_26-19_41_35 / P19_P20 2026_03_10-18_36_10 / P11_P12 (pilot) 2026_03_09-18_34_03 / P09_P10 2026_02_26-19_47_35 / P01_P02 (pilot)	2026_03_17-22_42_37 / P15_P16	2026_04_07-22_33_07 / P39_P40 2026_04_07-17_29_26 / P33_P34 2026_03_31-21_32_11 / P27_P28 2026_03_31-18_39_54 / P25_P26 2026_03_27-18_34_14 / P23_P24 2026_03_26-18_28_36 / P17_P18 2026_03_03-23_45_18 / P07_P08 2026_03_03-22_32_43 / P05_P06 2026_03_03-19_31_30 / P03_P04 (pilot)

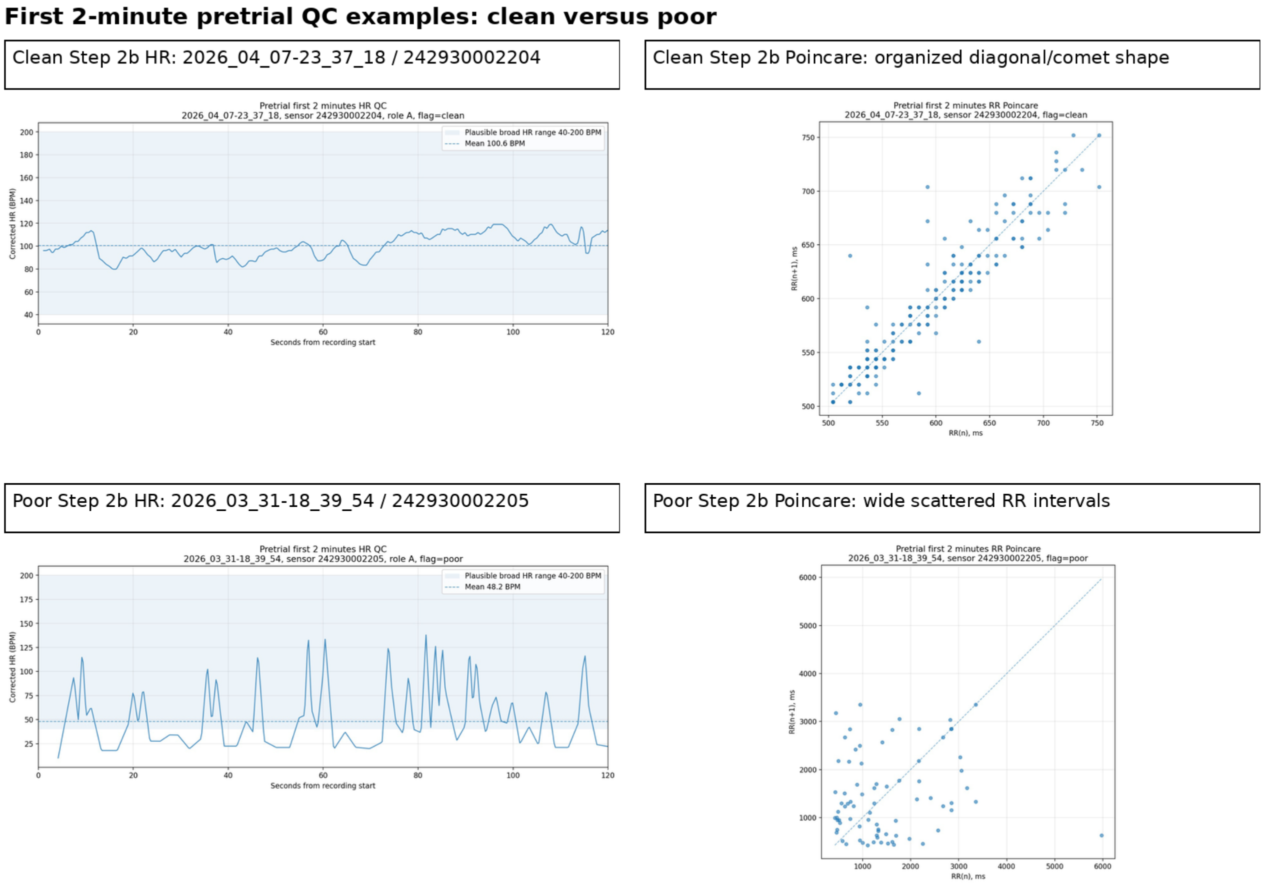


Figure 3. First 2-minute pretrial QC examples comparing clean and poor recordings. The clean panels use 2026_04_07-23_37_18 / sensor 242930002204 / role A. The poor panels use 2026_03_31-18_39_54 / sensor 242930002205 / role A. These examples show why Step 2b identifies baseline sensor-quality problems before task movement.

7. Step 3 - trial-level HR segmentation

Field	Content
Purpose	Cut full-session corrected HR into paired dyad-window files using table-motion timing windows. This creates the input for later correlation, PLI, and NSTE.
Script used	03_script_segment_hr_by_trial_EG.py
Pipeline origin	This is a TableTask-specific segmentation step. It uses outputs from the Lena-adapted Step 2, but the segmentation itself depends on the TableTask timing and condition structure.
Main inputs	qc_outputs/veronica_analysis_units.csv, processed_ecg_hr/hr_interpolated_4hz/, processed_ecg_hr/pretrial_2min_qc_v2/qc/
Main outputs	processed_ecg_hr/trial_segments_4hz/paired_trial_hr/, processed_ecg_hr/trial_segments_4hz/qc/03_trial_segmentation_summary.csv, 03_trial_segment_manifest.csv, 03_run_summary.txt
Run status	156/156 paired dyad-window segments succeeded. The script found all 36 corrected HR files.

Step 3 counts

- 156 total paired segments segmented successfully.
- 115 successful segments are main-analysis eligible at the metadata level.
- Trial segment QC: 99 clean, 4 caution, 53 poor.
- Main-analysis eligible trial QC: 75 clean, 4 caution, 36 poor.
- Condition counts among segmented rows: FF = 70, FE = 69, practice = 17.
- Condition counts among main-analysis eligible rows: FF = 58, FE = 57.

Step 3 is not synchrony analysis

- No correlation, PLI, or NSTE should be interpreted from Step 3.
- Step 3 only aligns HR_A and HR_B on a common 4 Hz time grid within each table-motion window.
- Step 3 carries forward condition labels, Step 1 exclusion fields, Step 2b pretrial QC flags, and trial-segment QC flags.

8. QC inclusion/exclusion logic

The flowchart below separates coverage exclusions, QC labels, and final-analysis eligibility. Poor segments are retained in files for documentation; they are not deleted at Step 3. [Update after final QC]

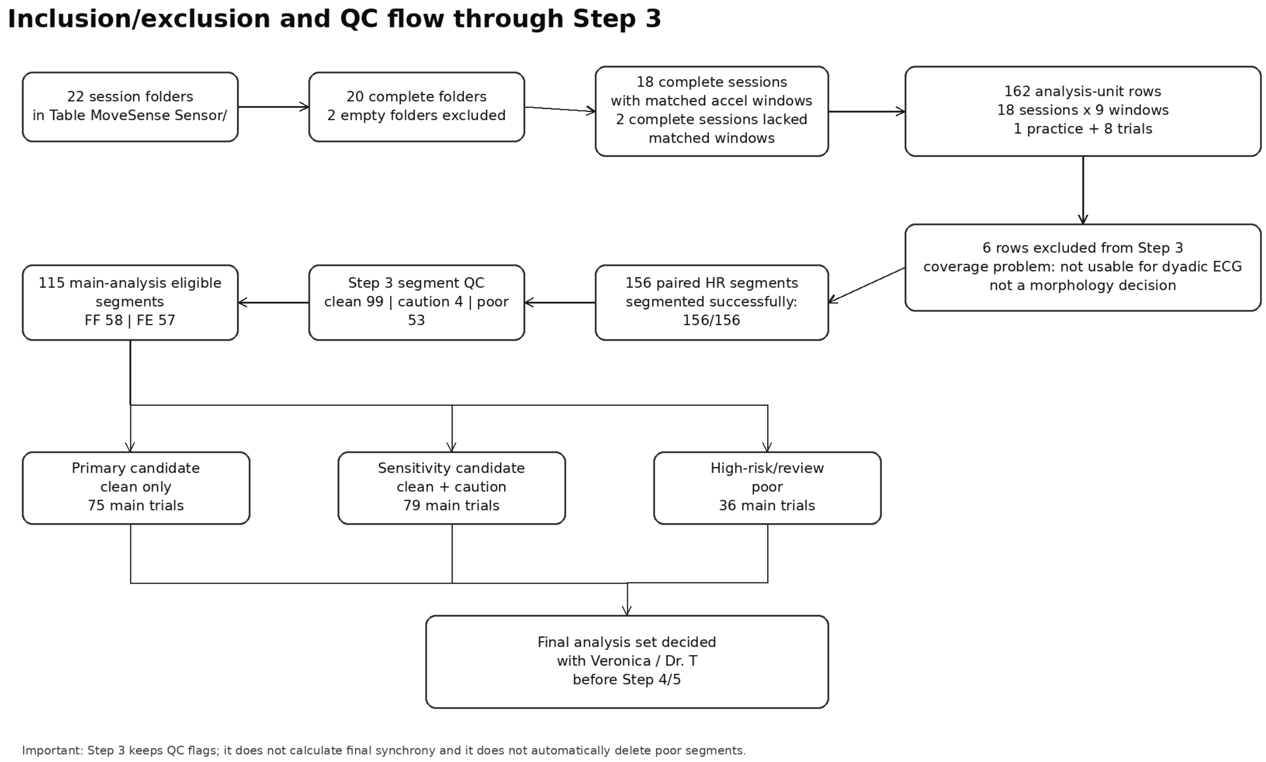


Figure 4. QC inclusion/exclusion logic through Step 3. Coverage exclusion, Step 2b pretrial QC, trial-segment QC, and final-analysis eligibility are kept separate. The primary analysis candidate is main-analysis eligible and clean; clean plus caution is reserved for sensitivity analysis after review.

Recommended decision rule before Step 4

Category	Recommended handling
Both clean / clean trial segment	Proceed to Step 4 primary synchrony analysis candidate.
Caution	Keep for sensitivity analysis or inspect manually before inclusion.
Poor	Retain for documentation, but exclude from primary synchrony/PLI/NSTE unless specifically approved after review.
Pilot	Keep for QC and training examples only. Exclude from main analysis.
Practice	Keep for documentation and QC. Exclude from final task-trial analysis.